

EXPERIMENT: Prolog Program for ADB with Name and DOB

Aim

To write and implement a Prolog program to store and retrieve ADB details using Name and Date of Birth (DOB).

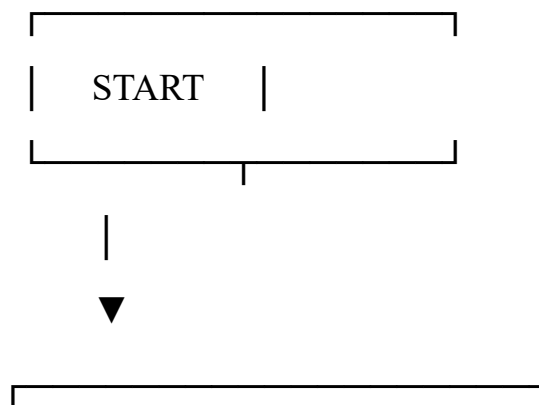
Objective

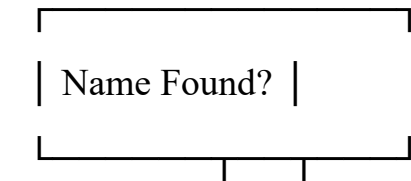
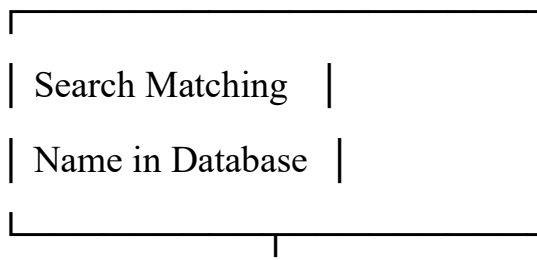
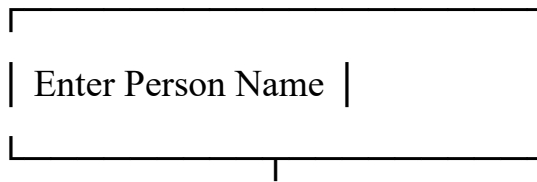
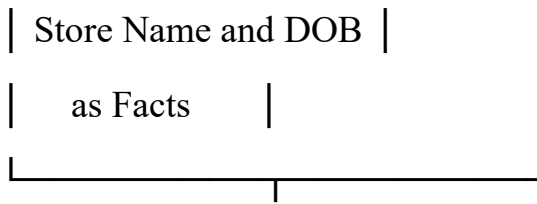
- To understand facts and rules in Prolog.
- To store a person's name and date of birth.
- To retrieve DOB information using a person's name.
- To demonstrate querying in Prolog.

Algorithm

1. Start the program.
2. Define facts containing the person's name and DOB.
3. Define a rule to retrieve the DOB using the person's name.
4. Enter a person's name as a query.
5. Match the name with the stored facts.
6. Display the corresponding DOB.
7. Stop the program.

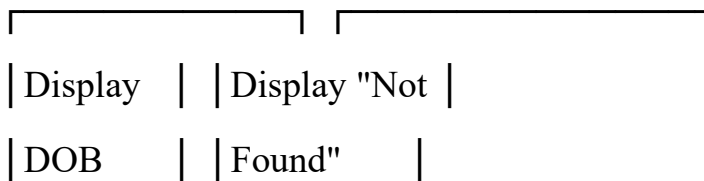
Flowchart

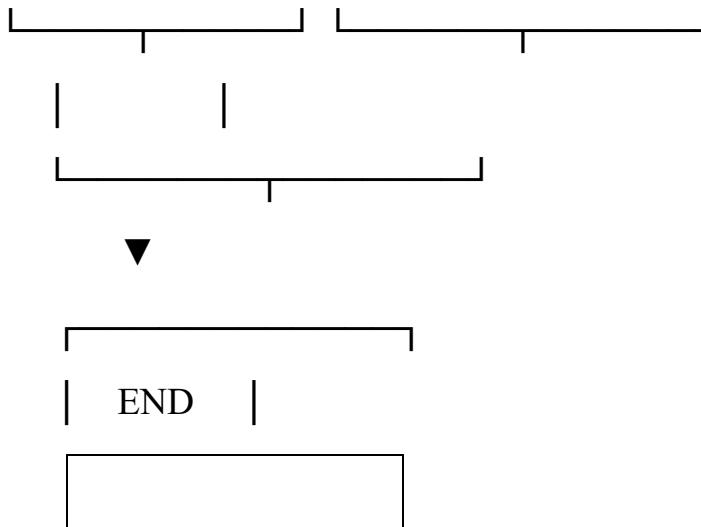




| |

Yes | | No





Code

```
database = {  
    "Arshad": "01/07/2005",  
    "Rahul": "20/08/2002",  
    "Priya": "10/12/2004",  
    "Anu": "25/03/2003",  
    "Kiran": "07/11/2001"  
}  
  
def display_all():  
    print("\n" + "=" * 45)  
    print("    ALL NAME AND DOB RECORDS")  
    print("=" * 45)  
    if len(database) == 0:  
        print("Database is empty.")  
        return  
    for name, dob in database.items():  
        print("Name :", name)  
        print("DOB  :", dob)
```

```
    print("-" * 45)

def search_person():
    name = input("Enter name to search: ")
    if name in database:
        print("\nPerson Found!")
        print("Name :", name)
        print("DOB :", database[name])
    else:
        print("\nPerson not found.")

def add_person():
    name = input("Enter name: ")
    if name in database:
        print("Person already exists.")
        return
    dob = input("Enter DOB (DD/MM/YYYY): ")
    database[name] = dob
    print("Record added successfully.")

def update_person():
    name = input("Enter name to update: ")
    if name in database:
        new_dob = input("Enter new DOB (DD/MM/YYYY): ")
        database[name] = new_dob
        print("Record updated successfully.")
    else:
        print("Person not found.")
```

```
def delete_person():  
    name = input("Enter name to delete: ")  
    if name in database:  
        del database[name]  
        print("Record deleted successfully.")  
    else:  
        print("Person not found.")  
  
def main():  
    while True:  
        print("\n")  
        print("=" * 45)  
        print("      ADB - NAME AND DOB DATABASE")  
        print("=" * 45)  
        print("1. Display All Records")  
        print("2. Search Person")  
        print("3. Add Person")  
        print("4. Update DOB")  
        print("5. Delete Person")  
        print("6. Exit")  
        choice = input("Enter your choice: ")  
        if choice == "1":  
            display_all()  
        elif choice == "2":  
            search_person()
```

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elif choice == "3":
    add_person()
elif choice == "4":
    update_person()
elif choice == "5":
    delete_person()
elif choice == "6":
    print("\nProgram terminated.")
    break
else:
    print("\nInvalid choice. Please try again.")
if __name__ == "__main__":
    main()
```

Output:



```
*IDLE Shell 3.12.5*
File Edit Shell Debug Options Window Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/shaik arshad basha/AppData/Local/Programs/Python/Python312/code.py

=====
          ADB - NAME AND DOB DATABASE
=====
1. Display All Records
2. Search Person
3. Add Person
4. Update DOB
5. Delete Person
6. Exit
Enter your choice: 2
Enter name to search: Arshad

Person Found!
Name : Arshad
DOB : 01/07/2005

=====
          ADB - NAME AND DOB DATABASE
=====
1. Display All Records
2. Search Person
3. Add Person
4. Update DOB
5. Delete Person
6. Exit
Enter your choice: |
```

Ln: 33 Col: 19

Result

The Prolog program was successfully implemented to store and retrieve a person's Date of Birth using their name.

Conclusion

The program demonstrates how facts and rules can be used in Prolog to create a simple database containing names and DOBs. The required DOB can be retrieved by providing the person's name as a query.